

TICK-LEVEL ANALYSIS

1. THE STRUCTURE OF TICK DATA

1.1. Terminology. The basic important elements of market state change are trades, bids and offers. We use the term *tick data* to denote high-resolution observations of these numbers, and usually assume it is available as of any given time in which the market is open. The trades are collectively considered to comprise the *trade tape* while the bids and offers collectively form the centralized limit *order book* or *CLOB* as of any given moment.

If the market is a centralized exchange¹, we can consider one or more orders to be at the *top of book*. For typical exchanges this is the set of highest-price bids and lowest-priced offers that have not yet *matched* or been *cancelled*, and have not *traded out*. The top of book is normally also ranked with highest priority to the oldest orders, in which case the exchange is said to use classic *price-time priority*. Incoming new orders are said to comprise the *order flow*.

For a given asset class, the group of exchanges serving it, the rules they use, and the firms accessing them collectively form the *market microstructure*.

1.1.1. Data Formats. Incoming tick data from established fully electronic markets normally arrives in FIX format². Take, for example, data³ from the CME such as

```
1128=9SOR9=265SOR35=XSOR49=CMESOR34=10065111SOR52=20141013152659076SOR75=20141013
SOR268=2SOR279=2SOR22=8SOR48=656784SOR83=215750SOR107=ESV4 C1895
SOR269=1SOR270=1775SOR271=5SOR273=152659000SOR336=2SOR346=1SOR1023=1
SOR279=0SOR22=8SOR48=656784SOR83=215751SOR107=ESV4 C1895SOR269=1
SOR270=1850SOR271=291SOR273=152659000SOR336=2SOR346=6SOR1023=3SOR10=008SOR
```

Less established markets may use JSON-style formats as in this message from the Coinbase cryptocurrency exchange

```
{'type': 'open', 'side': 'sell', 'price': '0.03743000',
  'order_id': '88d50b31-d5c6-445b-9134-05989d05e165', 'remaining_size': '0.01000000',
  'product_id': 'ETH-BTC', 'sequence': 1601762096, 'time': '2019-02-18T19:42:34.700000Z',
  'received_utc': 1550518954.092663}
```

1.1.2. A Note On Timestamps. It is common, and desirable, to represent timestamps using UTC, except for human presentations. However, note that there have been over 15 billion seconds since 1970. Therefore the 15-digit resolution of IEEE double precision floating point arithmetic will track these timestamps to slightly worse than the microsecond level.

1.1.3. Building The Book. Market data feeds, historical and otherwise, arrive in FIX (or sometimes JSON) packets in *delta format*, which is to say incoming packets modify an existing picture of the entire order book. Periodic transmission of an *order book snapshot* allow new entrants to form an overall picture of orders, and the subsequent delta updates keep it accurate.

¹Modern decentralized exchanges have an alternative quoting structure.

²See [QuickFix](#) for an open source parser, and [FixWiki](#) for open source documentation. More official documentation is available from [Fiximate](#).

³Here the notation **SOR** indicates the ASCII Start-Of-Record character.

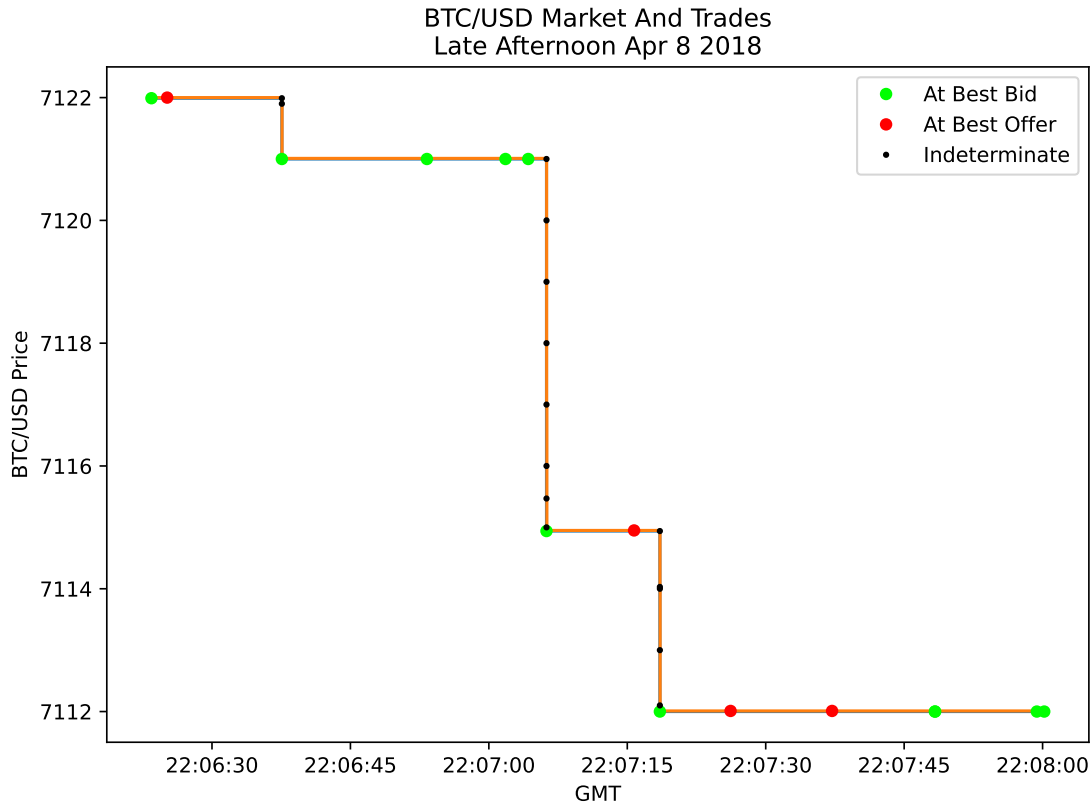


FIGURE 1. Bitcoin market and trades over a two minute period. Bid/offer spread is not visible, but we see the market undergoing multiple level changes. Note the three cases of large sell orders crossing the spread, taking out several levels of resting buy orders.

The incoming delta updates may be level 3 data, which provides an identifier for each incoming order packet, including price, quantity, side and perhaps other flags. Multiple orders at any given price level are aggregated to the overall quantity at that price level.

In contrast level 2 data does not bother with individual orders, instead providing quantity updates at each price level without specifying what orders correspond to the update.

1.1.4. *Inferring Queue Position.* For someone sending resting orders into the book, hoping to have a new entity hit their bid or pay their offer, knowledge of *queue position* is very important. Directionless market noise will touch the orders in the top queue position, yielding half-spread profits every time. Queue positions at the back of the line, in contrast, are associated with someone blowing through the book level and moving the price.

1.2. **Trades Come From Orders.** One of the important things to keep in mind when analyzing market data is that any reported trade exhausted the *smaller* associated order. So, a large market order crossing and (say) hitting 5 orders on the best bid price will result in 5 trade reports, of which at least 4 can be assumed to exhaust an entire order quantity. Usually, this will have the same exchange-reported timestamp for each of the 5 trades, and the trade

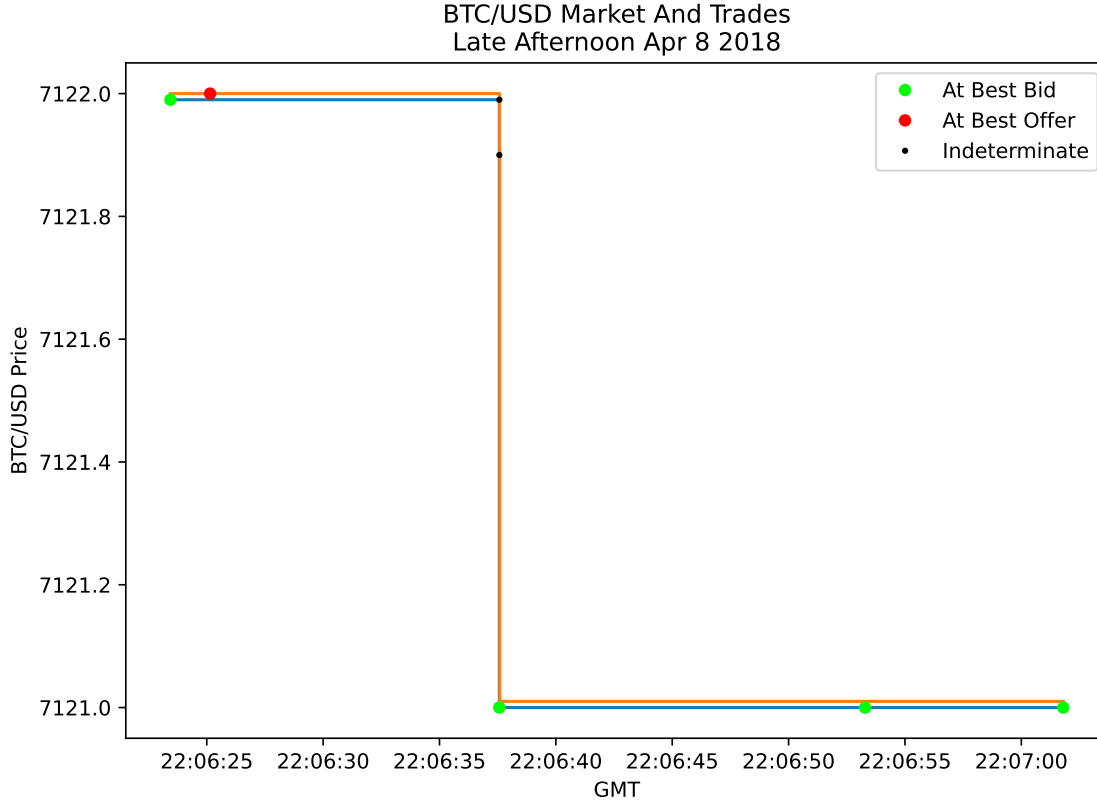


FIGURE 2. Bitcoin market and trades over a two minute period. Bid/offer spread is somewhat visible.

reports are additionally likely (but not certain) to arrive simultaneously in the same network packet.

1.3. Intermarket Lead-Lag. Lead-lag relationships between two markets, either on the same asset or similar assets, are prevalent in the world’s electronic markets. They are not usually unidirectional over macroscopic periods of time, but generally speaking the more liquid market will have more *information discovery* and a corresponding lead over the less liquid market.

Among the more robust ways to identify lead-lag is to examine market behavior near “large” trades, such as the 98th percentile in one market. Say this is asset A.. With accurately-recorded timestamps, it is usually possible to detect lagging trades in (say) asset B or, failing that, at least in book updates for asset B.

Though a regression is applicable in these cases, it is usually best to look at actual timestamps, to develop a sense of the competitive landscape in terms of systems and network speeds.

2. INTERESTING ALTERNATIVE MARKET STRUCTURES

The U.S. equity markets are a *fractured market*, in the sense that the same equities are traded on multiple exchanges. They are now legally linked by *Reg NMS* (i.e. *national market*

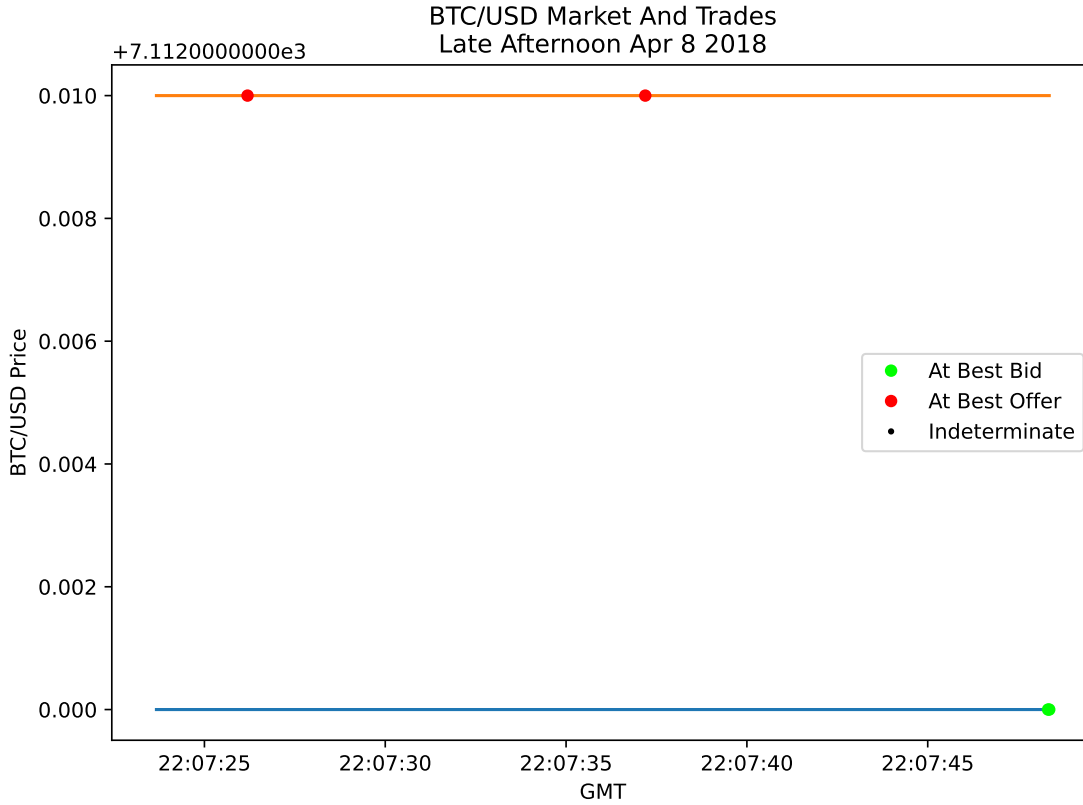


FIGURE 3. Bitcoin market and trades over a 20 second period. Bid/offer spread is obvious, but the market had no level changes.

structure) which specifies rules under which they must send orders to each other when prices are better-looking for the customer.

These regulatory NMS processes are slow compared to private networks, yielding some opportunity for cross-exchange (statistical) arbitrage.

2.1. Fractured US Equity Market Structure. The fractured US equity markets admit a variety of fee structures, technology, order types, and listings choices. Because the equities traded are fully fungible, the assets traded at these exchanges are exactly the same. This is unlike, for example, futures contracts on otherwise similar commodities, which are technically distinct assets.

2.2. Multi-market Trading. Most US equity market participants direct orders to more than one exchange, in the interest of getting more trading done more quickly, and amortizing the technological costs of developing trading systems.

2.2.1. Fee Structures. One of the keenest areas of competition between exchanges lies in fee structures. Exchanges collect fees on every trade, and sometimes have fees for orders sent as well. These fees are a primary source of income for exchanges⁴. Setting fees high increases per-trade revenue, but causes market participants to steer more for their orders to the competition,

⁴Other significant sources of exchange income include data access fees, and colocation rents.

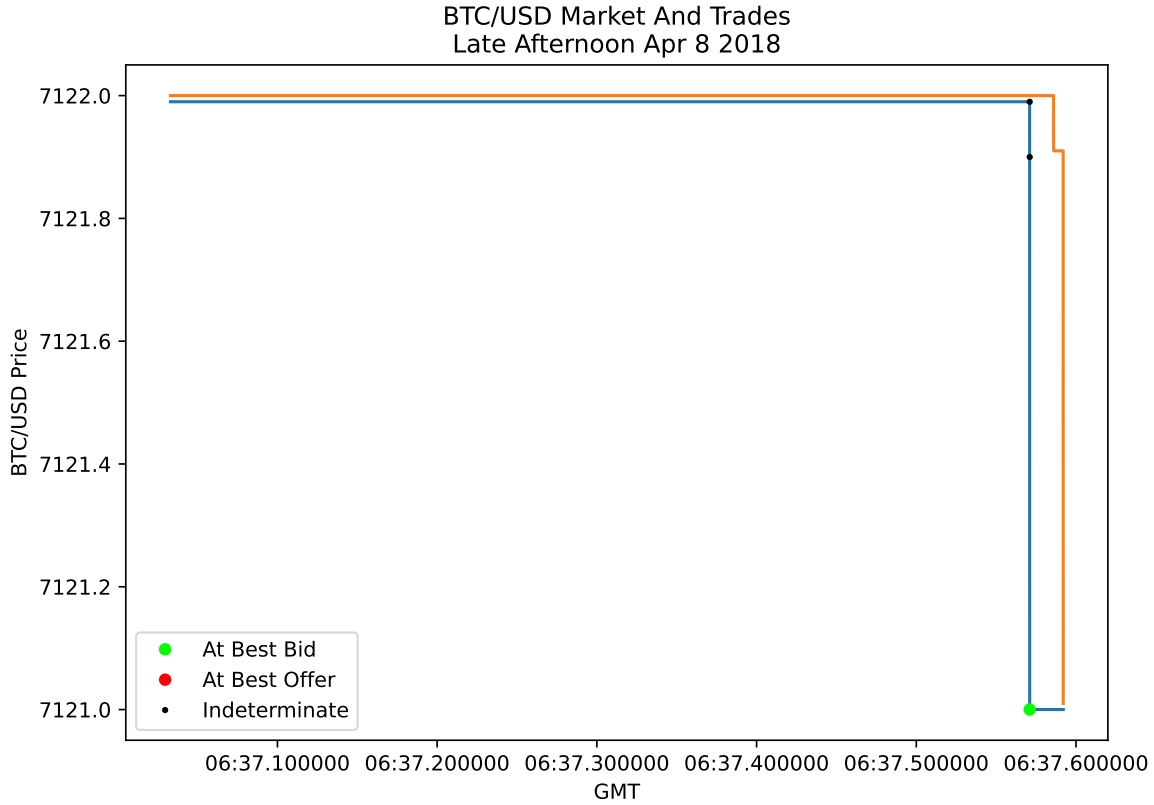


FIGURE 4. Bitcoin market and trades over a 1 second period. We observe the market trading through the bid.

reducing trade volume. For small-time clients, the exchange does not necessarily mind, but high-volume clients are usually entered into low-fee tiers in order to attract and retain their business.

Market makers provide a *service* to the markets, in the form of placing orders that other can choose or decline to interact with. In partial compensation for this, exchanges typically have low trade fees, or even rebates, applied to the *passive* or market-maker, side of a trade, and high trade fees applied to the *aggressive* or market-taker, side.

Some exchanges (e.g. Nasdaq OMX BX) take on *inverted* fee structures, in which the market-taker side gets a rebate and the market-maker side pays more. In these exchanges, market-makers typically widen spreads a bit. Their orders are likelier to attract interest and activity superior to the levels they otherwise might attract, potentially compensating for the fees.

Not all exchanges make the orders visible to participants. These *dark pools*, common for US equities, match orders without displaying any kind of book to market participants, which in theory allow participants to place large orders, and make trades at lower market impact costs. They frequently have minimum size rules for matching. In practice accumulation algorithms have tended to outperform dark pools for that purpose.

2.3. Other Alternative Market Structures. Foreign exchange *electronic crossing networks (ECN)* allow brokers *last look rights*. Futures exchanges may use a combination of price-time priority and *proportional allocation* in which all orders at the top-of-book get a nonzero fill probability.

Some exchanges allow brokers to give higher priority to *internal crosses* that join the broker's own customers. The U.S. equity markets now have similar structure, in effect, as orders arising from *retail* order flow are internally crossed by Citadel and other firms who make it their business to purchase the flow, cross where sensible, and then send the *exhaust* as new orders to the usual markets.

Certain futures markets are traded *pro rata*, in which there is no time priority given, or time priority is given only to the very earliest order. In particular the extremely liquid EuroDollar contract at the Chicago Mercantile Exchange trades on a *modified version of pro rata* allocation.

Eric Budish suggests discretizing not only price, in order price ticks, but also times, by *altering the matching process* so as to hold periodic auctions.

3. TICK LEVEL STATISTICS

3.1. Basic Book Statistics. The most important tick-level statistic is the *mid price*, defined as

$$\text{mid} = \frac{1}{2}(\text{bid} + \text{offer})$$

Next most important is the *VWAP* or *volume-weighted average price* defined for trades of sizes V_i at prices P_i to be

$$\text{VWAP} = \frac{\sum_1^N P_i V_i}{\sum_1^N V_i}$$

The VWAP is a minimal representation of tradable position changes.

Somewhat less useful is a *TWAP* or *time-weighted average price*, which performs the same sum over time periods rather than trades.

More interesting is an *order-weighted average price*, formed from the observation that if we have negative *book pressure* in the form many more offers than bids, the *clearing price* at which participants collectively would be willing to trade were tick sizes smaller is closer to the bid than the mid price. This is not an exact science, but given a top of book with N_B bids at price B and N_A offers at price A one reasonable formula weights with square roots

$$w = \sqrt{N_A} + \sqrt{N_B}$$

$$\text{OWA} = \frac{\sqrt{N_A}}{w} B + \frac{\sqrt{N_B}}{w} A$$

Clearly, other weighting schemes, possibly involving further levels of the book, can be incorporated into alternative versions of the OWA.

Arguably, the OWA is a predictive model, since we can use it to predict whether the next trade will occur on the bid side or the offer side of the book. The economic intuition is that if an order book has a great deal of orders on the offer, and very few on the bid, then (A) quite a few people are trying to sell and (B) it would not take many of them crossing the book to eliminate all the remaining bid orders, thereby dropping it one level. The trouble with OWA

as a predictive model is that its predictions are strong precisely when many orders are already in the books on one side, ahead of any orders placed by a us as a new order-placer.

3.2. Trade Marking. A very common derived statistic is *marked* trades, where bids and offers are marked according to whether a buyer initiated the transaction, or the seller initiated it. This process is flattered as the *Lee-Ready Algorithm* and essentially consists of the following for each trade

- Find the best bid and offer as of the trade time
- Compare the trade price to them
- If the trade price equals the best bid, mark as seller-initiated
- If the trade price equals the best offer, mark as buyer-initiated
- (Optional) Otherwise, mark according to whether trade price was below or above mid price

As stated, the algorithm may be too simple. For example block trades in equity markets and trades on quoted calendar spreads in futures markets may not be appropriate for marking.

Easley *et alia*'s *VPIN* or *volume-weighted probability of informed trading* for a time t compares buyer-initiated versus seller-initiated trades in N windows preceding t . The idea is that if many trades are crossing the spread in the same direction, then the market is likely to have more informed directional buyers than *noise traders* who lack particular information.

We begin by defining *order imbalance* in the n th window by taking V^B to be the volume of buyer-initiated trades and V^S to be the volume of seller-initiated trades, and setting the imbalance as

$$I_n = \frac{|V_n^B - V_n^S|}{V_n^B + V_n^S}$$

We can now set VPIN to be the mean

$$\text{VPIN} = \frac{1}{N} \sum_{n=1}^N I_n$$

3.3. Flow. A common (and simpler) extension of this idea is *trade flow*, which is a running tally of signed trade sizes where the sign is defined as 1 if the trade was seller-initiated and -1 if it was buyer-initiated. At any moment, we examine all reported trades within the last time period of length τ . Using the notation above, we can then define flow as

$$F_t^{(\tau)} = V_{(t-\tau, t)}^B - V_{(t-\tau, t)}^S$$

Clearly this can be normalized or transformed in many ways, with the goal of achieving more predictive forms.

The essential idea behind flow as a quantitative metric is that, in circumstances when many sellers are willing to cross that market-making bid-offer spread to complete their transactions, there is likely to be new information driving their choices. We do not know exactly what it is, but we certainly want to adapt to it.

When flow has a stable sign over macroscopic periods of time, it is a manifestation of supply and demand imbalance, and so it (usually) corresponds to steady price movements over a given time period. In a price time series, this has the appearance of *momentum* but the fundamental underlying process is economic (rather than the mysticism of momentum).

3.3.1. *Order Update Flow.* Just as we can examine the flow of actual trades, we can also examine the *potential* upcoming trades in the form of incoming *order flow*. Here, we relate a large number or total quantity of incoming buy orders to likely price increases, and vice versa for sell orders.

Because order flow is a well-know driver of price change, and yet orders have little inherent cost to send, it is possible to manipulate markets by placing orders solely for the purpose of generating a reaction. This is quite illegal in most advanced traditional finance, but tolerated in other large markets such as centralized crypto exchanges.

3.4. **Returns.** We commonly think of the interval T return of an asset as of time t in either absolute terms, as

$$r_t^{(T)} = \frac{P_t}{P_{t-T}} - 1$$

or in logarithmic terms

$$r_t^{(T)} = \log \left(\frac{P_t}{P_{t-T}} \right)$$

but we now see that the idea of knowing a single price P_t at some time t is somewhat optimistic. Really all we have is a set of order prices, and perhaps a recent trade price to go by.

Let's say we were to define P_t as the mid price. Naturally this stays constant over some time intervals, implying that returns are constantly decreasing in absolute value over this time. In addition, the return in any period becomes fairly sensitive to exactly which ticks the period contains.

We cannot entirely overcome the ambiguity, but we can certainly alleviate the tick-sensitivity problem by using an EWMA. We define

$$r_t^{(T)} = \log \left(\frac{P_t}{\text{EWMA} \left(P, \lambda = \frac{1}{T} \right)} \right)$$

which can conveniently be computed without saving past ticks.

4. THE EPPS EFFECT

A common issue with tick level data is the long periods of boredom interrupted by short intervals of terror. That is, data updates come at certain times (often bunched) so at a sufficiently small timescale correlations appear to drop to zero.

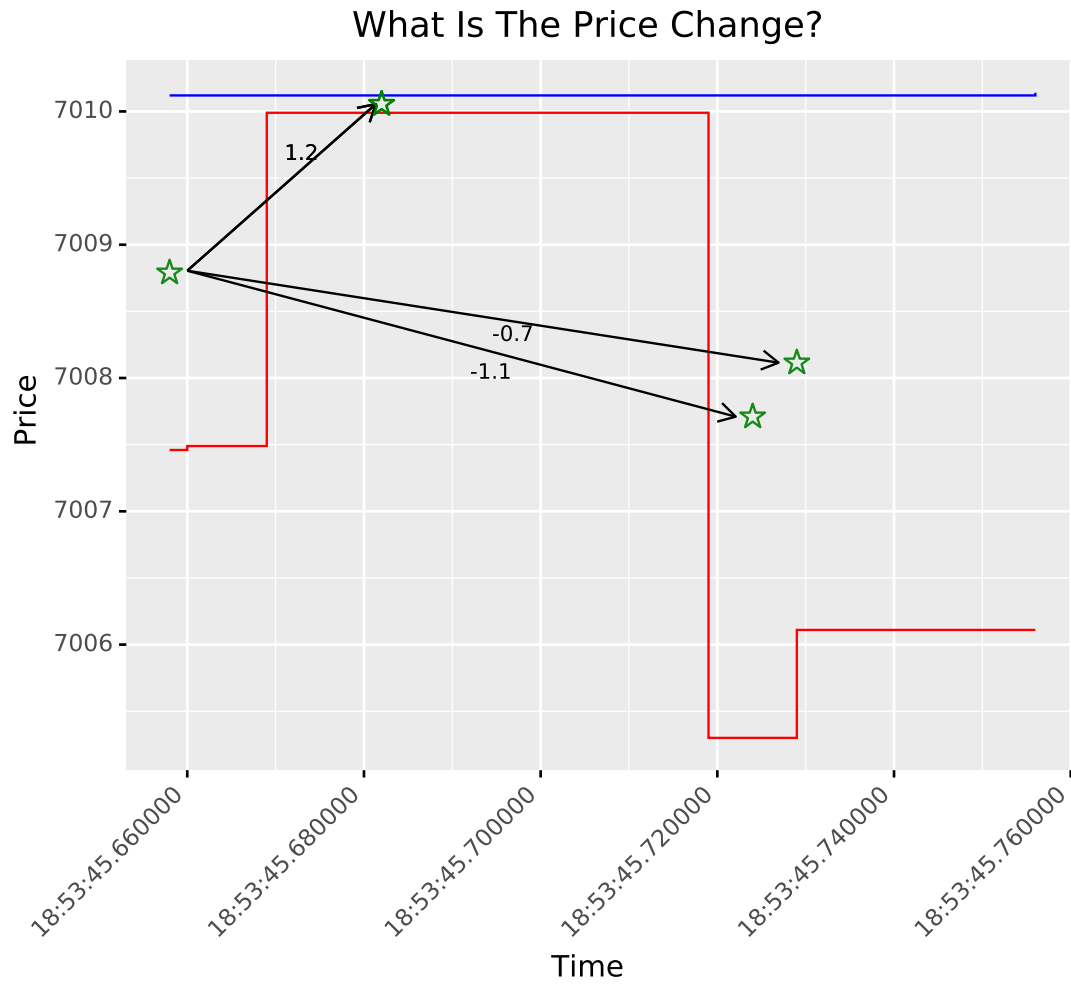


FIGURE 5. Price changes and returns are ambiguous in definition, and appear to vary wildly, at short time scales.